PRISM: A Pedagogical Resource Inspection System for Multimodal Educational Content Quality Evaluation

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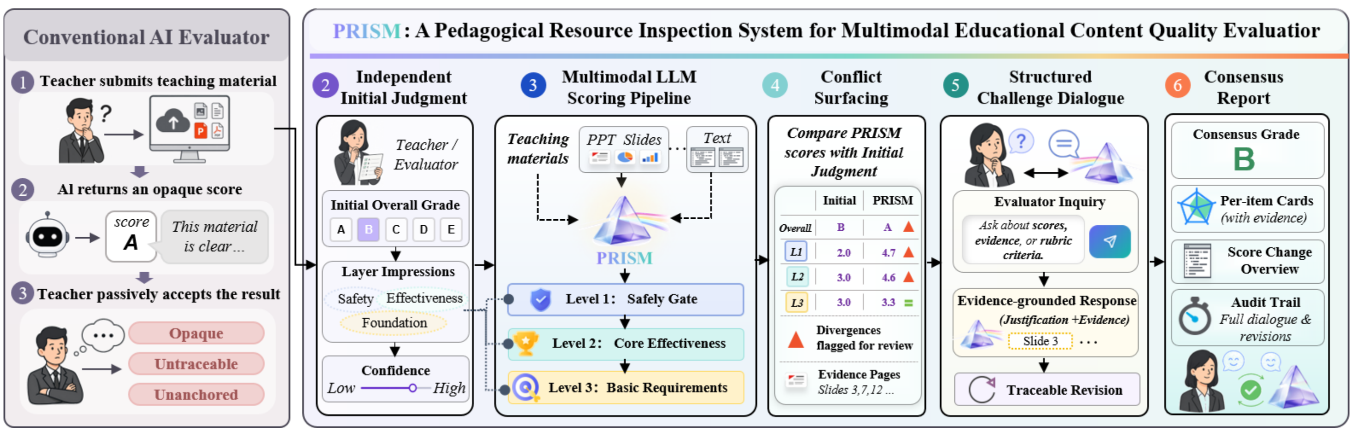

Figure 1: System overview. A conventional AI evaluator (left) returns an opaque score the teacher passively accepts; PRISM (right) scaffolds a five-stage workflow: independent initial judgment, multimodal LLM scoring against a three-layer non-compensatory rubric, conflict surfacing, structured challenge dialogue, and an audited consensus report.

Abstract

We present **PRISM** (**P**edagogical **R**esource **I**nspection **S**ystem for **M**ultimodal content), an interactive system for evaluating AI-generated and teacher-created multimodal educational resources. PRISM grounds evaluation in a validated 24-indicator three-layer non-compensatory rubric: a safety gate (L1) with veto authority, a core effectiveness layer (L2) yielding a five-tier grade (A to E), and a diagnostic layer (L3). A multimodal LLM jointly processes slide images and text with page-level evidence.Beyond "show-and-accept" scoring, PRISM supports traceable human–AI challenge and revision. Evaluators record an *independent initial judgment* before AI scores are revealed and may contest any score through traceable dialogue.

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CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → *Education.*

Keywords

educational resource evaluation; multimodal LLM; human-in-the-loop; challenge mechanism; non-compensatory rubric

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1 Introduction

The rapid adoption of generative AI in education has produced a paradox: teachers can now create slide decks and lesson plans in minutes, yet the quality of AI-generated materials varies widely in content accuracy, pedagogical soundness, and ethical compliance [9]. Applying a general-purpose LLM as evaluator produces holistic judgments that conflate safety-critical failures (fabricated citations, undisclosed AI authorship) with mild pedagogical weaknesses; rubric-guided LLM scoring improves alignment with human judges but does not enforce safety asymmetry [3, 4]. Neither approach addresses the multimodal nature of modern teaching materials, where image-text consistency and media quality require joint visual-linguistic reasoning that recent multimodal LLM research has begun to address for documents [5, 11] and unified image-text understanding [6].

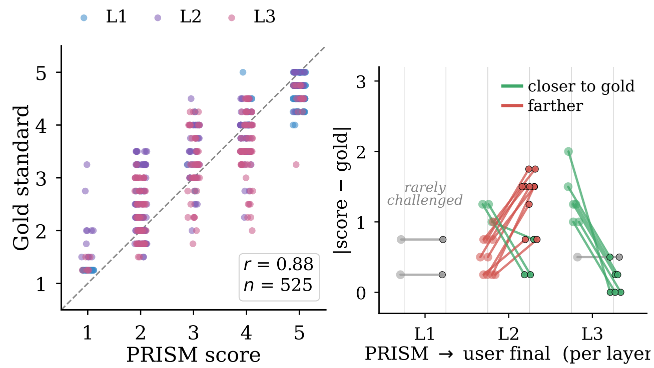
PRISM contributes **(1)** a theory-grounded 24-indicator rubric that enforces a non-compensatory veto so safety-critical failures cannot be offset by pedagogical strengths; **(2)** a multimodal LLM scoring engine that reasons jointly over slide images and text with page-level evidence; and **(3)** a five-dimensional challenge mechanism, building on proactive conversational AI [2], in which an *independent initial judgment* seeds a structured-challenge dialogue, operationalising appropriate-reliance principles for educational-resource evaluation (Figure 1).

2 Evaluation Framework

PRISM organises evaluation along a 24-indicator three-layer non-compensatory rubric (Figure 1), grounded in Merrill's First Principles [8], Mayer's CTML [7], the ICAP framework [1], and UNESCO AI-in-education guidelines [9], and validated through a Modified Delphi study with 15 domain experts. **L1 Safety Gate** (6 indicators covering factual accuracy, ethics, privacy, link safety, citation verifiability, and AI-authorship disclosure) screens for content correctness and ethical compliance: any L1 score ≤ 2 triggers an immediate "prohibited from use" verdict regardless of L2/L3 performance, operationalising the fatal-flaw principle [10]. **L2 Core Effectiveness** (11 indicators on teaching aims, content coverage, learning activities, cognitive design, and assessment) is the grade-determining layer, with the indicator mean mapped to grades A/B/C/D/E (thresholds 4.5 / 3.5 / 2.5 / 1.5). **L3 Basic Requirements** (7 indicators on visual design, readability, media quality, metacognitive support, and technical usability) contributes diagnostic detail without affecting the grade. This non-compensatory layering is the key architectural decision distinguishing PRISM from holistic LLM scoring, which has no equivalent veto.

3 System Design

Architecture and Workflow. PRISM is implemented as a Next.js front end with a Python/FastAPI back end. The five-step workflow proceeds as: *(i)* resource upload with metadata, used to conditionally activate indicators; *(ii) independent initial judgment* (overall A to E grade, three-layer impressions, and a confidence rating) recorded *before* any AI output is shown; *(iii)* three sequential LLM rounds, one per layer, with real-time progress; *(iv)* a comparison view that surfaces conflicts and supports the challenge phase (Section 3); *(v)* a final report combining overall-grade cards, a three-column qualitative summary (strengths, weaknesses, suggestions), and per-indicator detail panels with embedded challenge records.



(a) Indicator-level agreement (b) Layered challenge effect

Figure 2: Pilot evaluation by rubric layer. *(a)* PRISM vs. 4 experts gold-standard scores. *(b)* Challenge-induced shifts from PRISM to user-final scores.

Multimodal LLM Scoring. Each evaluation round receives *both* slide images and extracted text, since indicators on multimodal consistency, visual quality, and readability cannot be judged from text alone. Three sequential LLM calls (one per layer) each receive layer-specific rubric prompts with 1 to 5 behavioral anchors per indicator and return structured JSON (`code`, `score`, `comment`, `evidence_pages`). Separating layers keeps each layer independently auditable.

Five-Dimensional Challenge Mechanism. After AI scoring, PRISM enters a challenge phase organised along five dimensions (Figure 1): **(1) Conflict Surfacing** highlights indicators where AI and initial scores diverge by ≥ 1 grade level or ≥ 1.5 points; **(2) Challenge Modality** offers structured and free-form challenge; **(3) Response Strategy** requires every revision to cite specific slide pages and rubric anchors, with no concession on social pressure alone; **(4) Authority Locus** preserves the original PRISM score alongside the user-final judgment for traceable audit; and **(5) Learning Loop** retains challenge history across sessions, making recurring revision targets traceable across sessions.

4 Preliminary Evaluation

We conducted a pilot study with N=12 educators (3 in-service teachers, 9 graduate students), each evaluating two stage-matched resources from a curated 12-resource subset covering 5 quality tiers (A to E) and 8 subject domains. The gold standard was set by two parallel 4-expert panels with combined ICC[2,k] = 0.919. PRISM matched the gold standard exactly on 62.5% of sessions and within ±1 grade tier on all of them; per-indicator MAE was 0.643 (Pearson r=0.88, $p < .001$). Validity varies systematically across layers (L1: 0.380; L2: 0.676; L3: 0.869). The 22 challenge-induced revisions diverge sharply across layers (Figure 2): L3 revisions are uniformly corrective (7/7 closer to gold), L2 revisions skew toward user over-strictness (9/12 farther), and L1 is essentially unchallenged. SUS averaged 82.92 (Good on Bangor's scale) and TAM subscales 5.96 to 6.08 on a 7-point scale.

Acknowledgments

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